

Notes: Jorion, Liu, and Shi (JFE, 2005)
Informational Effects of Regulation FD: Evidence from Rating Agencies

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1 Big picture

This paper studies whether **Regulation Fair Disclosure (Reg FD)** changed the information role of credit rating agencies. Reg FD, implemented on October 23, 2000, prohibited public firms from selectively disclosing material nonpublic information to favored market professionals, especially

equity analysts. But Reg FD explicitly exempted credit rating agencies, so firms could still provide confidential information to Moody's, S&P, and Fitch for the purpose of producing public ratings.

The central idea is simple. If equity analysts lost access to selective disclosure while rating agencies retained it, then rating changes should become more informative after Reg FD. The authors test this by examining the stock market reaction to credit rating changes before and after Reg FD.

Main finding

The stock price reaction to rating changes becomes much stronger after Reg FD:

- **Downgrades** become more negative: the raw three-day abnormal return falls from about -4.57% before Reg FD to -6.93% after Reg FD.
- **Upgrades** become meaningfully positive: the raw three-day abnormal return rises from an insignificant -0.11% before Reg FD to a significant $+1.42\%$ after Reg FD.
- The effects are stronger for firms that likely had more pre-FD selective disclosure: larger firms and firms with more analyst following.

Interpretation: Reg FD did not simply remove selective information from the market. It shifted the relative informational advantage toward credit rating agencies, turning rating changes into more powerful public signals.

2 Incremental contribution of the paper

1. **New angle on Reg FD.** Most Reg FD research asks whether equity analysts became less informed. This paper asks whether a Reg FD-exempt intermediary - credit rating agencies - became more informative.
2. **Institutional contribution.** The paper highlights an important exemption in the regulation: firms may still disclose nonpublic information to rating agencies, as long as ratings are made public and the information is used for rating purposes.
3. **Empirical contribution to the rating-agency literature.** Prior work generally finds that downgrades affect stock prices but upgrades have little or no impact. This paper shows that after Reg FD, upgrades also generate a significant positive stock price reaction.
4. **Policy contribution.** The paper shows that disclosure regulation can change the allocation of informational rents across information intermediaries, not merely the total amount of information in markets.
5. **Identification contribution.** The authors combine a pre/post event-study design with cross-sectional tests based on firm size and analyst following, which are proxies for the extent of pre-FD selective disclosure.

3 Institutional background

3.1 What Reg FD changed

Before Reg FD, firms could selectively share material nonpublic information with favored investors, market professionals, and analysts. Reg FD required that if firms intentionally disclosed material

nonpublic information to such groups, they had to disclose it simultaneously to the public.

3.2 Why rating agencies are different

Credit rating agencies were exempted from Reg FD. Rating agencies therefore continued to receive confidential information such as internal forecasts, capital spending plans, and major corporate event information. This institutional feature creates the paper’s testable prediction: **rating changes should become more informative after Reg FD because rating agencies remain a channel for selective information while equity analysts lose access to it.**

3.3 Rating scale

The paper converts categorical credit ratings into a cardinal scale. AAA/Aaa is coded as 1, progressively worse ratings receive larger numbers, and default is coded as 23. This cardinal scale lets the authors measure the size of a rating change with a simple variable, *RCHANGE*.

4 Data and measurement

4.1 Sample

The paper uses rating changes from the Mergent Fixed Investment Securities Database (FISD). The sample includes taxable corporate bonds issued by U.S. firms.

- **Pre-FD period:** August 1998 to September 2000.
- **Post-FD period:** November 2000 to December 2002.
- October 2000 is excluded because Reg FD became effective on October 23, 2000.
- Final sample: **1,767 downgrades** and **437 upgrades**.

4.2 Event-study outcome

The main outcome is the stock price reaction around a rating change. The event window is three trading days: $(-1, 0, +1)$, where day 0 is the effective date of the rating change.

The raw cumulative abnormal return is

$$CAR_j = \sum_{t=-1}^{+1} (R_{jt} - R_{mt}),$$

where R_{jt} is the stock return of issuer j and R_{mt} is the value-weighted market return.

The standardized abnormal return is

$$SCAR_j = \frac{CAR_j}{\sigma(R_j - R_m)\sqrt{3}}.$$

Standardization matters because firms differ substantially in stock return volatility.

4.3 Contaminated versus non-contaminated events

Some rating changes occur near other firm-specific news. The authors use the *Wall Street Journal* Abstract to flag events as contaminated if price-relevant firm news appears within a three-day window around the rating change. They show that the main findings survive after excluding contaminated observations.

5 Empirical design

5.1 Baseline cross-sectional regression

The main regression is estimated separately for downgrades and upgrades:

$$SCAR_j = \alpha_0 + \alpha_1 RFD_j + \alpha_2 RCHANGE_j + \alpha_3 IGRADE_j + \alpha_4 DAYS_j + \varepsilon_j.$$

- RFD_j : equals one after Reg FD and zero before Reg FD.
- $RCHANGE_j$: absolute magnitude of the rating change on the cardinal scale.
- $IGRADE_j$: equals one if the rating change crosses the investment-grade boundary.
- $DAYS_j$: log number of days since the prior rating change in the same direction.

The coefficient of interest is α_1 . If rating agencies become more informative after Reg FD, α_1 should be negative for downgrades and positive for upgrades.

5.2 Sensitivity and cross-sectional tests

The paper then asks whether the pre/post result could be confounded by the 2001 recession. It also asks whether the effect is stronger where Reg FD should matter more:

- Large firms versus small firms.
- High analyst-following firms versus low analyst-following firms.

The prediction is that post-FD rating changes should be especially informative for large firms and firms with more analysts, because these firms were more likely to have had active pre-FD selective disclosure channels.

6 Tables and figures

6.1 Rating scale and sample construction

Table 1
Classification by credit ratings

Explanation	Standard & Poor's (modifiers)	Moody's (modifiers)	Fitch (modifiers)	Cardinal scale
<i>Investment grade</i>				
Highest grade	AAA	Aaa	AAA	1
High grade	AA (+, none, -)	Aa (1,2,3)	AA (+, none, -)	2, 3, 4
Upper medium grade	A (+, none, -)	A (1,2,3)	A (+, none, -)	5, 6, 7
Medium grade	BBB(+, none, -)	Baa (1,2,3)	BBB(+, none, -)	8, 9, 10
<i>Speculative grade</i>				
Lower medium grade	BB (+, none, -)	Ba (1,2,3)	BB (+, none, -)	11, 12, 13
Speculative	B (+, none, -)	B (1,2,3)	B (+, none, -)	14, 15, 16
Poor standing	CCC (+, none, -)	Caa (1,2,3)	CCC	17, 18, 19
Highly speculative	CC	Ca	CC	20
Lowest quality, no interest	C	C	C	21
In default	D		DDD/DD/D	23

Table 1. Credit rating categories and cardinal scale.

Read:

- Table 1 maps the agencies' letter ratings into a cardinal scale. Better ratings have lower values; worse ratings and default have higher values.
- Table 2 shows that the sample contains many more downgrades than upgrades. The pre-FD period has 777 downgrades and 310 upgrades; the post-FD period has 990 downgrades and 127 upgrades.
- The average size of rating changes is similar before and after Reg FD, which helps rule out the possibility that larger post-FD reactions are mechanically caused by larger rating changes.
- Contamination rates are also similar before and after Reg FD.

Economic message: The sample is suitable for a pre/post comparison because the average magnitude and contamination of rating changes do not shift dramatically around Reg FD. The main difference after Reg FD is therefore not that rating agencies make mechanically larger rating changes, but that the market reacts more strongly to similar rating changes.

Table 2

Year/Period	Downgrades		Upgrades		
	Number	Size	Number	Size	
<i>Panel A: distribution of number and size of bond rating changes by year and period</i>					
1998	[151]	1.70	[70]	1.20	
1999	[373]	1.73	[142]	1.21	
2000	[351]	1.57	[113]	1.41	
2001	[528]	1.76	[86]	1.28	
2002	[364]	1.65	[26]	1.92	
Pre-FD	[777]	1.68	[310]	1.29	
Post-FD	[990]	1.70	[127]	1.38	
Total	[1,767]	1.69	[437]	1.32	
Period	Downgrades		Upgrades		
	Number	%	Number	%	
<i>Panel B: percentage of contaminated rating changes pre- and post-FD</i>					
Pre-FD	All sample	[777]	100	[310]	100
	Contaminated	[172]	22.14	[45]	14.52
Post-FD	All sample	[990]	100	[127]	100
	Contaminated	[245]	24.75	[24]	18.90
Diff. in post- and pre-contamination rates (Z-statistic)			2.61 (1.28)	4.38 (1.14)	

The whole sample consists of 1,767 downgrades and 437 upgrades of taxable corporate bonds issued by U.S. firms during the period of 52 months from August 1998 to December 2002 excluding October 2000. The sample is split equally into two 26-month periods: the pre-FD period from August 1998 to September 2000 and the post-FD period from November 2000 to December 2002. Categorical bond ratings are converted into a cardinal variable measured on a 23-point scale (1 for rating AAA, 23 for rating D). Different gradations of a major category (e.g., BB+, BB, and BB-) are treated separately and assigned different integer points. Size of rating changes is the cardinal value of the new rating minus the cardinal value of the old rating. The number of rating changes is indicated in square brackets. Contaminated rating changes are identified by searching the Wall Street Journal Abstract using firm name. An observation is considered as contaminated if any firm-specific price-relevant information appears in the Wall Street Journal within a three-day window surrounding the event day of rating change. Z-statistic is used to test the difference in post- and pre-contamination rates.

Table 2. Number, size, and contamination of rating changes.

6.2 Rating-change categories and raw event-study results

Table 3

Summary of rating changes by absolute magnitude of rating changes, within class, across class, and across investment grade

Absolute magnitude of rating changes	Downgrades				Upgrades			
	Pre-FD		Post-FD		Pre-FD		Post-FD	
	#	%	#	%	#	%	#	%
<i>Panel A: sample distribution by absolute magnitude of rating changes</i>								
1	458	58.94	550	55.56	251	80.97	100	78.74
2	206	26.51	277	27.98	39	12.58	15	11.81
3	67	8.62	108	10.91	15	4.84	9	7.09
4	28	3.60	41	4.14	2	0.65	1	0.79
5	6	0.77	6	0.61	2	0.65		
6	4	0.51	5	0.51				
7	4	0.51					2	1.57
8	1	0.13						
9	2	0.26	1	0.10	1	0.32		
10			1	0.10				
11	1	0.13	1	0.10				
Total	777	100.00	990	100.00	310	100.00	127	100.00
Rating change	Downgrades				Upgrades			
	Pre-FD		Post-FD		Pre-FD		Post-FD	
	#	%	#	%	#	%	#	%
<i>Panel B: sample distribution by within class, across class, and across investment grade</i>								
Total	777	100.00	990	100.00	310	100.00	127	100.00
Within class	345	44.40	446	45.05	186	60.00	73	57.48
Across class	432	55.60	544	54.95	124	40.00	54	42.52
Across invest. grade	78	10.04	112	11.31	41	13.23	18	14.17

The sample consists of 1,767 downgrades and 437 upgrades of taxable corporate bonds issued by US firms during the period of 52 months from August 1998 to December 2002 excluding October 2000. The sample is split equally into two 26-month periods: the pre-FD period from August 1998 to September 2000 and the post-FD period from November 2000 to December 2002. Categorical bond ratings are converted into a cardinal variable measured on a 23-point scale (1 for rating AAA, 23 for rating D). Different gradations of a major category (e.g., BB+, BB, and BB-) are treated separately and assigned different integer points. The Magnitude of rating changes is the cardinal value of the new rating minus the cardinal value of the old rating. Within or across class indicates whether the rating change occurs within gradations of the same

Table 3. Distribution by magnitude, rating class, and investment-grade boundary.

Read:

- Table 3 shows that most rating changes are small. A one-notch change accounts for about 56–59 percent of downgrades and about 79–81 percent of upgrades.
- Downgrades are more likely than upgrades to cross rating classes. Cross-investment-grade changes are relatively uncommon but potentially important because many investors face investment-grade restrictions.
- Table 4 gives the main univariate result. Downgrade CARs become more negative after Reg FD, moving from -4.57% to -6.93% . The post-minus-pre difference is -2.37% .
- Upgrade CARs become positive and significant after Reg FD, moving from -0.11% before Reg FD to $+1.42\%$ after Reg FD.
- The same qualitative results appear in the non-contaminated sample and in standardized CARs.

Economic message: Table 4 is the clearest evidence that the information content of ratings changes rose after Reg FD. The upgrade result is especially important because prior studies often found no meaningful market reaction to upgrades. After Reg FD, even favorable rating news contains information that equity markets appear to value.

Table 4

The stock price response to bond downgrades and upgrades for the pre- and post-FD periods

	Downgrades			Upgrades		
	Mean % CAR	Median % CAR	% of CAR < 0	Mean % CAR	Median % CAR	% of CAR > 0
<i>Panel A: raw CAR</i>						
<i>All observations</i>						
Pre-FD	-4.57 (-7.76)***	-2.41	64.74	-0.11 (-0.34)	-0.47	46.13
Post-FD	-6.93 (-10.79)***	-2.14	61.62	1.42 (2.68)***	0.63	56.69
Difference (Post-Pre)	-2.37 (-2.65)***	0.27 (1.10)	-3.12	1.53 (2.53)***	1.11 (2.51)***	10.56
<i>Non-contaminated observations</i>						
Pre-FD	-3.06 (-5.52)***	-2.02	64.52	-0.22 (-0.70)	-0.49	46.27
Post-FD	-4.85 (-7.27)***	-1.38	58.27	1.17 (2.09)**	0.63	57.28
Difference (Post-Pre)	-1.79 (-2.06)**	0.64 (0.06)	-6.25	1.39 (2.24)**	1.12 (2.19)**	11.01
	Downgrades			Upgrades		
	Mean % SCAR ^a	Median % SCAR	% of SCAR < 0	Mean % SCAR	Median % SCAR	% of SCAR > 0
<i>Panel B: standardized CAR</i>						
<i>All observations</i>						
Pre-FD	-0.4039 (-7.87)***	-0.3203	64.74	-0.0039 (-0.07)	-0.0749	46.13
Post-FD	-0.7429 (-10.75)***	-0.3398	61.62	0.2432 (2.88)**	0.1257	56.69
Difference (Post-Pre)	-0.3390 (-3.94)***	-0.0195 (1.45)	-3.12	0.2471 (2.40)**	0.2006 (2.46)**	10.56
<i>Non-contaminated observations</i>						
Pre-FD	-0.2857 (-5.91)***	-0.2581	64.52	-0.0141 (-0.25)	-0.0720	46.27
Post-FD	-0.4773 (-7.12)***	-0.2212	58.27	0.2102 (2.48)**	0.1257	57.28
Difference (Post-Pre)	-0.1917 (-2.22)**	0.0369 (0.04)	-6.25	0.2243 (2.13)**	0.1977 (2.25)**	11.01

The whole sample consists of 1,767 downgrades and 437 upgrades of taxable corporate bonds issued by US firms during the period of 52 months from August 1998 to December 2002 excluding October 2000. The sample is split equally into two 26-month periods: the pre-FD period from August 1998 to September 2000 and the post-FD period from November 2000 to December 2002. *** and ** indicate significance better than the 1% and 5% two-tailed levels, respectively. Contaminated rating changes are identified by searching the Wall Street Journal Abstract using firm name. An observation is considered as contaminated if any firm-specific price-relevant information appears in the Wall Street Journal within a three-day window surrounding the event day of rating change. CAR is the cumulative abnormal return defined as the stock return minus the contemporaneous return on the value-weighted market portfolio, calculated over the three-day event window $(-1, +1)$, where day 0 is the effective date of a rating change. SCAR is the standardized cumulative abnormal return, defined as the ratio of CAR to its standard error calculated over the whole sample period August 1998 to December 2002. *T*-statistics are given in parentheses below the mean % CAR and SCAR. Mean and median differences are tested using *T* and Wilcoxon two-sample tests respectively.

Table 4. Stock price response to rating changes before and after Reg FD.

6.3 Multivariate event-study regressions and business-cycle check

Table 5
The effect of FD on the stock price response to bond downgrades and upgrades
 $SCAR_j = \alpha_0 + \alpha_1 RFD_j + \alpha_2 RCHANGE_j + \alpha_3 IGRADE_j + \alpha_4 DAYS_j + \varepsilon_j$

Independent variables	Downgrades		Upgrades	
	Expected sign	Coefficients (t-stat.)	Expected sign	Coefficients (t-stat.)
<i>Panel A: all observations</i>				
INTERCEPT		-0.1232 (-0.79)		0.0529 (0.29)
RFD	-	-0.3112 (-3.51)***	+	0.2510 (2.41)**
RCHANGE	-	-0.2743 (-6.42)***	+	-0.0311 (-0.54)
IGRADE	-	-0.47764 (-3.32)***	+	-0.1053 (-0.74)
DAYS	+/-	0.0467 (2.16)**	+/-	-0.0005 (-0.02)
Adj. R^2 (%)		4.78		0.64
F-stat.		23.16***		1.70
No. of obs.		1,767		437
Independent variables	Downgrades		Upgrades	
	Expected sign	Coefficients (t-stat.)	Expected sign	Coefficients (t-stat.)
<i>Panel B: noncontaminated observations</i>				
INTERCEPT		-0.0068 (-0.04)		0.0358 (0.20)
RFD	-	-0.1675 (-1.99)*	+	0.2269 (2.12)**
RCHANGE	-	-0.2404 (-5.59)***	+	-0.0348 (-0.62)
IGRADE	-	-0.2477 (-1.78)	+	-0.0732 (-0.49)
DAYS	+/-	0.0289 (1.38)	+/-	-0.0010 (-0.04)
Adj. R^2 (%)		3.43		0.33
F-stat.		12.98***		0.53
No. of obs.		1,350		368

The whole sample consists of 1,767 downgrades and 437 upgrades of taxable corporate bonds issued by US firms during the period of 52 months from August 1998 to December 2002 excluding October 2000. The sample is split equally into two 26-month periods: the pre-FD period from August 1998 to September 2000 and the post-FD period from November 2000 to December 2002. *** and ** indicate significance better than the 1% and 5% two-tailed levels, respectively. Contaminated rating changes are identified by searching the Wall Street Journal Abstract using firm name. An observation is considered as contaminated if any firm-specific price-relevant information appears in the Wall Street Journal within a three-day window surrounding the event day of rating change. SCAR is the standardized cumulative abnormal return, defined as the ratio of CAR to its standard error calculated over the whole sample period August 1998 to December 2002, where CAR is the cumulative abnormal return (daily stock return minus the contemporaneous return on the value-weighted market portfolio) over the three-day event window (-1, +1); RFD is a Regulation FD dummy variable equal to one if an observation is from the post-FD period and zero otherwise; RCHANGE is the absolute magnitude of the rating change, where categorical bond ratings are converted into a cardinal variable measured on a 23-point scale (1 for rating AAA, 23 for rating D); IGRADE is a dummy variable set equal to one if a bond is revised from investment grade to speculative grade or vice versa, and zero otherwise; DAYS is the natural log of the number of days between two consecutive rating changes of the same bond in the same direction, with the number of days set equal to 1,200 if there are no bond revisions in the same direction in the sample period.

Table 5. Regression evidence on the Reg FD effect.

Read:

- Table 5 confirms the univariate findings after controlling for the size of the rating change, whether the change crosses the investment-grade boundary, and the time since the previous change.
- For downgrades, the post-FD coefficient is -0.3112 in standardized abnormal return units and is statistically significant. Using the authors' approximate average three-day standard deviation of 4.5 percent, this corresponds to roughly -1.4% additional stock market reaction.
- For upgrades, the post-FD coefficient is +0.2510 and statistically significant.
- Table 6 checks whether the 2001 recession mechanically explains the post-FD finding. The business-cycle dummy moves the effects toward zero rather than away from zero, so the recession is unlikely to be the source of the stronger post-FD reactions.

Economic message: Table 5 shows that the post-FD effect survives controls and is not driven merely by larger rating changes. Table 6 strengthens the interpretation by showing that the recession does not explain the results. The policy change is therefore a more plausible explanation than the macroeconomic environment.

Table 6
Business cycle and the stock price response to bond downgrades and upgrades for post-FD observations: November 2000–December 2002
 $SCAR_j = \alpha_0 + \alpha_1 BCYCLE_j + \alpha_2 RCHANGE_j + \alpha_3 IGRADE_j + \alpha_4 DAYS_j + \varepsilon_j$

Independent variables	Downgrades		Upgrades	
	Expected sign	Coefficients (t-stat.)	Expected sign	Coefficients (t-stat.)
INTERCEPT		-0.4323 (-2.57)**		0.9074 (3.13)***
BCYCLE		0.5254 (3.82)***		-0.2473 (-1.22)
RCHANGE	-	-0.3439 (-5.10)***	+	-0.1673 (-1.80)*
IGRADE	-	-0.6195 (-2.86)**	+	0.0354 (0.14)
DAYS	+/-	0.0001 (0.65)	+/-	-0.0003 (-1.76)*
Adj. R^2 (%)		5.25		2.18
F-stat.		14.71***		1.70
No. of obs.		990		127

The sample consists of 990 downgrades and 127 upgrades of taxable corporate bonds issued by US firms during the post-FD period of 26 months from November 2000 to December 2002. ***, **, and * indicate significance better than the 1%, 5%, and 10% two-tailed levels, respectively. SCAR is the standardized cumulative abnormal return, defined as the ratio of CAR to its standard error calculated over the whole sample period August 1998 to December 2002, where CAR is the cumulative abnormal return (daily stock return minus the contemporaneous return on the value-weighted market portfolio) over the three-day event window (-1, +1); BCYCLE is a dummy variable equal to one if an observation is from the period November 2000 to November 2001, and zero otherwise; RCHANGE is the absolute magnitude of the rating change, where categorical bond ratings are converted into a cardinal variable measured on a 23 point scale (1 for rating AAA, 23 for rating D); IGRADE is a dummy variable set equal to one if a bond is revised downgraded from investment grade to non-investment grade or upgraded from non-investment to investment grade, and zero otherwise; DAYS is the natural log of the number of days between two consecutive rating changes of the same bond in the same direction, with the number of days set equal to 1,200 if there are no bond revisions in the same direction in the sample period.

Table 6. Business-cycle sensitivity check.

6.4 Cross-sectional evidence: size and analyst following

Table 7

The effect of Reg FD on the stock price response to bond downgrades and upgrades:

large firm vs. small firm $SCAR_i = \alpha_1' SIZE_{L_i} + \alpha_2' SIZE_{S_i} + \alpha_3' RFD * SIZE_{L_i} + \alpha_4' RFD * SIZE_{S_i} + \alpha_5' RCHANGE_i + \alpha_6' IGRADE_i + \alpha_7' DAYS_i + \epsilon_i$

large vs. small analyst following $SCAR_i = \alpha_1' AFL_{L_i} + \alpha_2' AFL_{S_i} + \alpha_3' RFD * AFL_{L_i} + \alpha_4' RFD * AFL_{S_i} + \alpha_5' RCHANGE_i + \alpha_6' IGRADE_i + \alpha_7' DAYS_i + \epsilon_i$

Independent variables	Downgrades			Upgrades		
	Expected sign	Coefficients (t-stat.)		Expected sign	Coefficients (t-stat.)	
		Size	Analyst		Size	Analyst
SIZE _L (AFL)		0.0477 (0.26)	0.009 (0.02)		-0.0084 (-0.05)	-0.0051 (-0.03)
SIZE _S (AFS)		-0.2232 (-1.35)	-0.105 (-1.14)		0.1143 (0.60)	0.1408 (0.73)
RFD*SIZE _L (AFL)	-	-0.5134 (-3.57)***	-0.5182 (-3.71)***	+	0.4688 (2.84)**	0.5151 (3.26)**
RFD*SIZE _S (AFS)	-	-0.2103 (-1.79)	-0.1652 (-1.42)	+	0.1057 (0.79)	0.0601 (0.44)
RCHANGE	-	-0.2651 (-6.14)***	-0.2709 (-6.31)***	+	-0.0208 (-0.50)	-0.0208 (-0.48)
IGRADE	-	-0.4773 (-3.31)***	-0.4718 (-3.28)***	+	-0.1260 (-0.88)	-0.1296 (-0.91)
DAYS	+/-	0.0469 (2.17)**	0.0469 (2.08)**	+/-	-0.0016 (-0.06)	-0.0037 (-0.14)
Adj. R ² (%)		13.37	13.37		2.69	3.08
F-stat.		39.95***	39.98***		1.70	1.95
No. of obs.		1,767		437		

The whole sample consists of 1,767 downgrades and 437 upgrades of taxable corporate bonds issued by US firms during the period of 52 months from August 1998 to December 2002 excluding October 2000. The sample is split equally into two 26-month periods: the pre-FD period from August 1998 to September 2000 and the post-FD period from November 2000 to December 2002. * and ** indicate significance better than the 1% and 5% two-tailed levels, respectively. SCAR is the standardized cumulative abnormal return, defined as the ratio of CAR to its standard error calculated over the whole sample period August 1998 to December 2002, where CAR is the cumulative abnormal return (daily stock return minus the contemporaneous return on the value-weighted market portfolio) over the three-day event window $(-1, +1)$; RFD is a Reg FD dummy variable equal to one if an observation is from the post-FD period, and zero otherwise; SIZE_L (SIZE_S) is a large (small) firm size dummy variable, set equal to one if a bond issuer's market value of equity at the end of fiscal year 1999 is greater (less) than the sample median of the variable, and zero otherwise; AFL (AFS) is a large (small) analyst following dummy variable set equal to one if a bond issuer has number of stock analysts following it at the end of fiscal year 1999 greater (less) than the sample median, and zero otherwise; RCHANGE is the absolute magnitude of the rating change, where categorical bond ratings are converted into a cardinal variable measured on a 23-point scale (1 for rating AAA, 23 for rating D); IGRADE is a dummy variable set equal to one if a bond is revised from investment grade to speculative grade or vice versa, and zero otherwise; DAYS is the natural log of the number of days between two consecutive rating changes of the same bond in the same direction, with the number of days set equal to 1,200 if there are no bond revisions in the same direction in the sample period.

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Table 7. Reg FD effects by firm size and analyst following.

Read:

- The post-FD downgrade effect is much stronger for large firms than for small firms: -0.5134 versus -0.2103 .
- The post-FD downgrade effect is also much stronger for high analyst-following firms than for low analyst-following firms: -0.5182 versus -0.1652 .
- For upgrades, the post-FD effect is also concentrated among large firms and firms with high analyst following.
- These cross-sectional differences line up with the idea that firms with more pre-FD analyst interaction were more affected by the loss of selective disclosure to equity analysts.

Economic message: Table 7 provides the most important support for the Reg FD interpretation. If the pre/post result were due only to a generic time-series change, there would be less reason for the effect to concentrate among firms with greater analyst following and larger market capitalization. The cross-sectional pattern instead points to the reallocation of private information from equity analysts toward rating agencies.

7 Key mechanisms and interpretation

7.1 Why downgrades matter more after Reg FD

Downgrades already mattered before Reg FD, consistent with prior studies. After Reg FD, they matter even more. The paper's interpretation is that rating agencies now embed information that other market intermediaries no longer receive through selective disclosure. A downgrade therefore becomes a stronger signal of unfavorable private information.

7.2 Why upgrades become significant after Reg FD

The upgrade result is conceptually important. Previous work often found that upgrades had little stock price effect, possibly because firms voluntarily disclosed good news themselves. After Reg FD, however, rating agencies may have relatively more privileged information even about favorable credit developments, causing upgrades to become significant signals.

7.3 Why firm size and analyst following matter

Large firms and high analyst-following firms likely had richer pre-FD private communication channels with equity analysts and institutions. When Reg FD constrained these channels but left rating agencies exempt, the information advantage of ratings should rise more for these firms. Table 7 is consistent with this prediction.

8 Limitations

- **Single policy date.** Like all Reg FD studies, the paper relies on one regulatory change, so unrelated time-series changes are difficult to rule out fully.
- **Recession overlap.** The post-FD period overlaps with the 2001 recession. The authors address this concern, but it remains an important institutional context.
- **Measurement of announcement dates.** The study uses FISD effective dates as event dates. Any measurement error should weaken the estimated market reaction.
- **Contamination risk.** Some rating changes coincide with other firm news. The non-contaminated sample helps, but not all information events can be perfectly observed.
- **Interpretation of stock reactions.** A stronger market reaction to ratings is consistent with more information content, but rating changes could also have real effects through borrowing costs or investment-grade constraints.

9 Short summary

Jorion, Liu, and Shi study whether Regulation FD increased the value of credit ratings as information signals. Because rating agencies were exempt from Reg FD, firms could still provide them with confidential information after selective disclosure to equity analysts was curtailed. Using an event study around rating changes, the authors show that both downgrades and upgrades have stronger stock price effects after Reg FD. The effect is stronger among large firms and firms with high analyst following, which are plausibly the firms most affected by the pre-FD selective-disclosure environment.

10 Conclusion

The paper's main lesson is that disclosure regulation can alter the information hierarchy among market intermediaries. Reg FD was intended to level the playing field by ending selective disclosure to favored market professionals. But because rating agencies were exempt, they became more important conduits of private information to the public. The stronger post-FD stock market reactions

to both downgrades and upgrades suggest that rating changes became more informative after the regulation.